

FedUA-Net: Calibrated Uncertainty-Aware Federated Learning for Privacy-Preserving Multi-Task Medical Imaging

Abstract—Federated learning (FL) enables multi-institutional healthcare collaborations without centralizing private patient imaging data. However, standard medical FL frameworks enforce task homogeneity—assuming all clinical centers solve identical diagnostic tasks. In multi-center clinical networks, institutions frequently deploy heterogeneous imaging modalities (e.g., MRI, Ultrasound, Digital Radiography) with mutually disjoint diagnostic categories (11 classes across 3 institutions). Single-task FL algorithms suffer severe degradation or cannot operate under this heterogeneity. To address this challenge, we present FedUA-Net (*Federated Uncertainty-Aware Network*), a multi-task cross-silo federated framework that learns an attention-augmented shared visual backbone (EfficientNetV2-S with CBAM) while strictly decoupling task-specific classification heads and Batch Normalization (BN) layers locally at each clinical site. To resolve representation skew in highly unbalanced multi-institutional consortia, we employ uniform server aggregation ($w_k = 1/K$), ensuring that data-rich clinical sites do not overpower specialized, data-scarce cohorts. Furthermore, to mitigate overconfident misclassifications in safety-critical triage, FedUA-Net introduces an end-to-end post-hoc uncertainty quantification and distribution-free conformal prediction pipeline using validation-guided Temperature Scaling and Adaptive Prediction Sets (APS). Evaluated across three clinical benchmarks—Brain Tumor MRI, Breast Ultrasound, and COVID-19 Chest X-Ray across three random seeds (mean $\pm$ std)—FedUA-Net achieves $93.30 \pm 1.13\%$ mean accuracy, $93.05 \pm 1.36\%$ macro-F1, and 0.894 ± 0.019 Matthews Correlation Coefficient, outperforming classical federated methods including FedAvg ($92.36 \pm 0.92\%$), FedBN ($92.34 \pm 0.99\%$), FedBABU ($91.99 \pm 0.35\%$), and FedProx ($91.97 \pm 1.03\%$). While isolated Local-Only (94.01%) and bi-level personalized Ditto (93.93%) achieve slightly higher uncalibrated point accuracy on isolated acoustic domains, they suffer from severe overconfidence (raw ECE 0.0463) and lack statistical safety guarantees. FedUA-Net deliberately navigates this trade-off by prioritizing clinical trustworthiness: post-hoc calibration reduces Expected Calibration Error (ECE) from $0.0504 \rightarrow 0.0307$, while conformal prediction delivers certified prediction sets (2.33 classes at 90% target coverage with 99.1% empirical coverage). Under severe local data scarcity ($N = 200$ and $N = 100$ images), FedUA-Net maintains strong cross-modality feature transfer over FedBN (+3.99% and +3.42% accuracy gains), establishing FedUA-Net as a reliable paradigm for collaborative healthcare AI.

Index Terms—Federated Learning, Medical Image Analysis, Multi-Task Learning, Uncertainty Quantification, Conformal Prediction, Temperature Scaling, Attention Mechanism.

I. INTRODUCTION

DEEP learning models have achieved remarkable success across diverse medical imaging modalities, including Magnetic Resonance Imaging (MRI), ultrasound, and digital radiography [1], [2]. However, training robust, clinically generalizable deep neural networks requires large, multi-institutional datasets. Centralizing medical images across clinical centers is severely restricted by patient privacy regulations (e.g., HIPAA in the USA, GDPR in the European Union), institutional data governance policies, and transmission bandwidth [3].

Federated Learning (FL) [4], [5] addresses these privacy concerns by enabling distributed clinical nodes to collaboratively optimize a global model by exchanging only model parameters or gradients, keeping raw patient data on local premise. Despite its promise, classical medical FL algorithms (e.g., FedAvg [4], FedProx [6], and FedBN [7]) operate under a restrictive assumption: **task homogeneity**. They assume every participating client solves an identical diagnostic task (e.g., all sites classifying chest radiographs into identical disease categories) and differs only in scanner manufacturer or patient demographics [8].

In real-world multi-center hospital networks, clinical collaborations are inherently heterogeneous across medical tasks and departments:

- 1) **Clinical Site A (Neurology Silo)**: Specializes in Brain Tumor T1-weighted contrast-enhanced MRI scans (4 classes: glioma, meningioma, pituitary tumor, normal) [9].
- 2) **Clinical Site B (Oncology Silo)**: Evaluates Breast Ultrasound (BUSI) scans (3 classes: benign, malignant, normal) [10].
- 3) **Clinical Site C (Pulmonology Silo)**: Screens Digital Chest X-Rays (4 classes: COVID-19, lung opacity, viral pneumonia, normal) [11].

Under standard single-task FL, these clinical departments cannot participate in collaborative learning because their input feature distributions and label spaces are mutually disjoint (11 distinct diagnostic classes across 3 clinical imaging modalities). Furthermore, in imbalanced networks where data-rich departments (e.g., pulmonary screening centers with 21,000 scans) coexist with specialized clinics (e.g., breast ultrasound clinics with 546 scans), classical sample-weighted averaging causes the dominant client to overwrite shared representations, inducing gradient starvation and representation skew on smaller sites [12].

Crucially, clinical deployment requires **trustworthy uncertainty quantification (UQ)** [13], [14]. Modern deep neural networks frequently produce overconfident

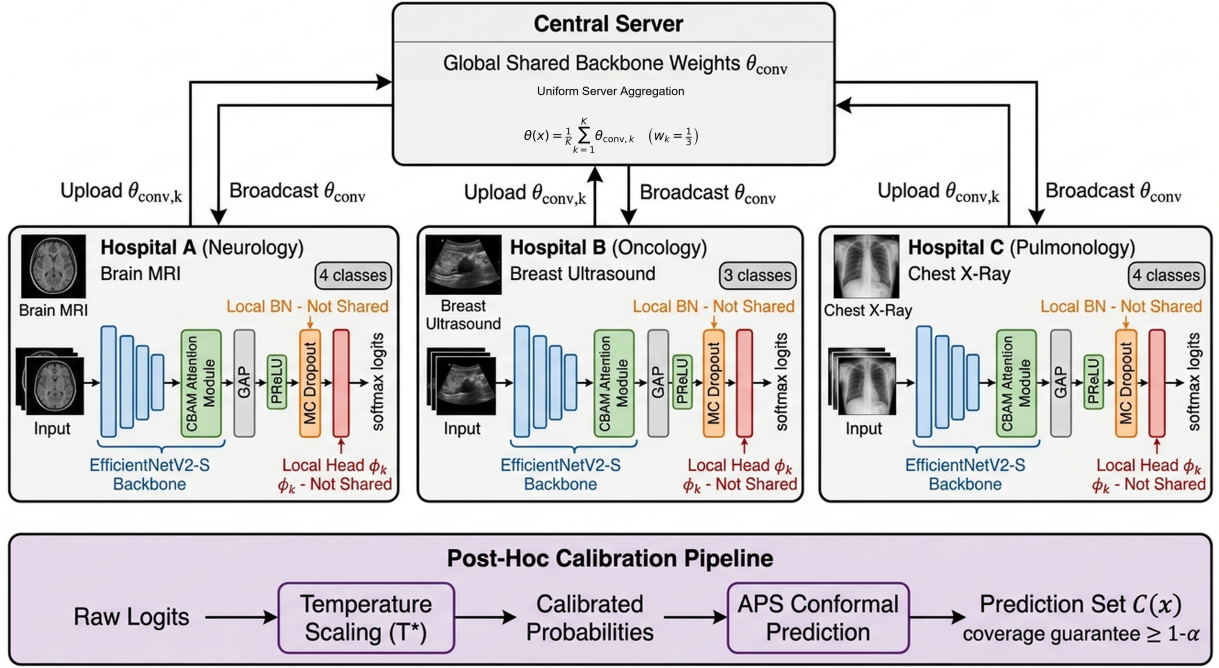


Fig. 1. Overview of the proposed **FedUA-Net** framework: Distributed clinical sites with heterogeneous imaging modalities and disjoint label spaces collaboratively train an attention-augmented shared visual backbone (EfficientNetV2-S with CBAM) while maintaining private Batch Normalization layers and local classification heads. Post-hoc Temperature Scaling and Adaptive Prediction Sets (APS) conformal prediction provide certified prediction sets with guaranteed local coverage.

misclassifications due to cross-entropy optimization and domain shift [15]. In safety-critical healthcare, an overconfident incorrect diagnosis can lead to delayed treatment or incorrect interventions. While isolated local models may obtain high empirical point accuracy on familiar training distributions, they lack rigorous calibration and statistical safety guarantees across heterogeneous cohorts. A clinically safe AI system must not only maximize predictive accuracy but also provide mathematically certified prediction sets: ambiguous or borderline cases must produce multi-label prediction sets with guaranteed statistical coverage, alerting radiologists for secondary evaluation [25], [26].

To resolve these challenges, we propose **FedUA-Net** (*Federated Uncertainty-Aware Network*), a unified framework for multi-task, multi-modal medical imaging with distribution-free conformal calibration (Fig. 1).

The principal contributions of this work are:

- **Multi-Task Federated Architecture with Decoupled Heads and Uniform Aggregation:** We propose an attention-augmented visual backbone (EfficientNetV2-S [16] with Convolutional Block Attention Modules (CBAM) [17]) combined with client-specific classification heads [18], [19] and local Batch Normalization (BN) layers [7]. The server aggregates visual representation layers with uniform client weighting ($w_k = 1/K$), enforcing an egalitarian fairness constraint across modalities while clients maintain private normalization statistics and task-specific classification heads.
- **End-to-End Uncertainty Recalibration Pipeline:** We integrate validation-guided Temperature Scal-

ing [15] to recalibrate predicted softmax logits, achieving substantial reductions in Expected Calibration Error (ECE) and Brier scores across all participating hospital nodes.

- **Distribution-Free Conformal Prediction with Exact Coverage Guarantees:** We implement Adaptive Prediction Sets (APS) split-conformal classification [26], providing provable local marginal coverage guarantees ($1 - \alpha$) on held-out patient cohorts without requiring distributional assumptions.
- **Rigorous Benchmark and Systematic Scarcity Evaluation:** Across three distinct clinical imaging modalities (Brain MRI, Breast Ultrasound, Chest X-Ray) over 3 random seeds, FedUA-Net achieves $93.30 \pm 1.13\%$ mean accuracy, outperforming classical federated baselines (FedAvg: 92.36%, FedBN: 92.34%, FedBABU: 91.99%, FedProx: 91.97%). On Hospital B, FedUA-Net achieves 88.60%, surpassing Centralized pooled training (88.03%). Under extreme local data scarcity ($N = 200$ and $N = 100$), FedUA-Net delivers consistent +3.99% and +3.42% accuracy advantages over FedBN.

II. RELATED WORK

A. Federated Learning in Medical Imaging

Federated learning was formulated by McMahan *et al.* [4] with Federated Averaging (FedAvg), aggregating model weights proportionally to local dataset size. In clinical settings, non-independent and identically distributed (Non-IID) data across hospitals presents substantial optimization

hurdles [5]. Li *et al.* [6] developed FedProx, adding a proximal regularization term to restrict local parameter drift. To handle scanner-induced domain shifts, Li *et al.* [7] proposed FedBN, demonstrating that localizing Batch Normalization layers while aggregating feature extraction weights significantly enhances cross-site domain adaptation. However, these classical methods assume homogeneous label spaces across all nodes.

B. Personalized and Multi-Task Federated Learning

To accommodate client heterogeneity, personalized FL (PFL) architectures have emerged. Arivazhagan *et al.* [18] introduced FedPer, separating deep networks into shared base layers and local personalization heads. Collins *et al.* [19] introduced FedRep, formally analyzing representation learning with local heads. Oh *et al.* [20] introduced FedBABU, freezing classification heads during federated optimization to learn generic body representations. Li *et al.* [21] proposed Ditto, formulating personalized FL as a bi-level optimization problem balancing local personalization with global consensus. FedUA-Net advances personalized federated learning by integrating dual-domain spatial-channel attention (CBAM) with post-hoc conformal calibration tailored for heterogeneous clinical tasks.

C. Uncertainty Quantification and Calibration

Modern deep neural networks suffer from severe miscalibration [15]. Guo *et al.* [15] demonstrated that post-hoc Temperature Scaling effectively aligns predicted confidence with empirical accuracy on validation splits without altering argmax classification predictions. Gal and Ghahramani [22] showed that Monte Carlo (MC) Dropout acts as a variational Bayesian approximation of Gaussian processes. Lakshminarayanan *et al.* [23] established deep ensembles for predictive uncertainty estimation, albeit with increased computational overhead. In FedUA-Net, we leverage feature-space Temperature Scaling and MC-Dropout to provide lightweight, highly calibrated confidence estimates suitable for edge hospital nodes.

D. Conformal Prediction in Healthcare

Conformal prediction, founded by Vovk *et al.* [24] and modernized by Angelopoulos and Bates [25], constructs prediction sets with exact finite-sample marginal coverage guarantees. Romano *et al.* [26] formulated Adaptive Prediction Sets (APS), accumulating sorted softmax probabilities until a calibrated threshold $\hat{q}$ is reached. While conformal prediction has shown promise in centralized medical diagnostics [27], its integration into multi-modal, multi-task cross-silo federated learning has remained largely unaddressed.

III. METHODOLOGY

A. Problem Formulation: Multi-Task Cross-Silo FL

Let K denote the number of participating hospital sites, where each site $k \in \{1, 2, \dots, K\}$ holds a private

local dataset $\mathcal{D}_k = \{(x_{k,i}, y_{k,i})\}_{i=1}^{N_k}$. The input space $\mathcal{X}_k$ represents site-specific medical imaging modalities (e.g., Brain MRI slices, Breast Ultrasound scans, or Chest X-Rays), and $\mathcal{Y}_k = \{1, 2, \dots, C_k\}$ denotes the discrete label space containing C_k mutually exclusive diagnostic categories. The label spaces across institutions are mutually disjoint:

$$\mathcal{Y}_j \cap \mathcal{Y}_k = \emptyset, \quad \forall j \neq k. \quad (1)$$

The total number of diagnostic categories across the entire healthcare consortium is $C_{\text{total}} = \sum_{k=1}^K C_k$. Patient data cannot leave the institutional boundary: $\mathcal{D}_k \cap \mathcal{D}_j = \emptyset$.

B. Decoupled Network Architecture

Each client k deploys a model $f_{\theta, \phi_k} : \mathcal{X}_k \rightarrow \mathbb{R}^{C_k}$ partitioned into two modular components:

- 1) **Shared Visual Backbone (g_θ):** A deep convolutional feature extractor parameterized by weights θ , shared across all clinical sites.
- 2) **Local Classification Head (h_{ϕ_k}):** A site-specific classification head parameterized by private weights $\phi_k \in \mathbb{R}^{D \times C_k}$, where $D = 512$ is the latent embedding dimension.

1) *Attention-Augmented Shared Feature Backbone:* The shared backbone g_θ consists of an EfficientNetV2-S architecture [16] pre-trained on ImageNet-1K, followed by a Convolutional Block Attention Module (CBAM) [17]. Given intermediate feature representation $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, CBAM sequentially applies 1D channel attention $\mathbf{M}_c \in \mathbb{R}^{C \times 1 \times 1}$ and 2D spatial attention $\mathbf{M}_s \in \mathbb{R}^{1 \times H \times W}$:

$$\mathbf{M}_c(\mathbf{F}) = \sigma(\text{MLP}(\text{AvgPool}(\mathbf{F})) + \text{MLP}(\text{MaxPool}(\mathbf{F}))), \quad (2)$$

$$\mathbf{F}' = \mathbf{M}_c(\mathbf{F}) \otimes \mathbf{F}, \quad (3)$$

$$\mathbf{M}_s(\mathbf{F}') = \sigma(f^{7 \times 7}([\text{AvgPool}(\mathbf{F}'); \text{MaxPool}(\mathbf{F}')])), \quad (4)$$

$$\mathbf{F}'' = \mathbf{M}_s(\mathbf{F}') \otimes \mathbf{F}', \quad (5)$$

where σ denotes the sigmoid function, $\otimes$ denotes element-wise multiplication, and $[\cdot; \cdot]$ denotes channel concatenation. Global average pooling (GAP) and a dense projection layer with PReLU activation map $\mathbf{F}''$ to latent vector $\mathbf{v} \in \mathbb{R}^D$.

2) *Local Batch Normalization Decoupling:* Let $\theta = \{\theta_{\text{conv}}, \theta_{\text{BN}}\}$, where θ_{conv} denotes convolutional and dense projection parameters, and $\theta_{\text{BN}} = \{\gamma, \beta, \mu_{\text{run}}, \sigma_{\text{run}}^2\}$ represents Batch Normalization affine scale/shift parameters and running statistics. To insulate the model against severe modality-induced feature shifts, all θ_{BN} are strictly excluded from federated aggregation and retained locally:

$$\theta_k^{(t)} = \{\theta_{\text{conv}}^{(t)}, \theta_{\text{BN},k}^{(t)}\}. \quad (6)$$

C. Federated Optimization Protocol

During communication round $t \in \{1, 2, \dots, T\}$:

- 1) **Server Broadcast:** The coordinator server broadcasts the global shared convolutional weights $\theta_{\text{conv}}^{(t)}$ to all active clinical nodes.
- 2) **Local Optimization:** Each client k initializes its body with $\theta_{\text{conv}}^{(t)}$, preserves its private $\theta_{\text{BN},k}$ and ϕ_k ,

and trains for E_k local epochs using balanced focal cross-entropy with label smoothing ($\epsilon = 0.1$):

$$\mathcal{L}_k = -\frac{1}{N_k} \sum_{i=1}^{N_k} w_{y_{k,i}} \log \hat{p}_{k,i}(y_{k,i}), \quad (7)$$

where class weights $w_c = \frac{N_k}{C_k \cdot N_{k,c}}$ balance local class frequencies.

- 3) **Uniform Server Aggregation:** Participating clients upload updated body parameters $\theta_{\text{conv},k}^{(t+1)}$. In contrast to standard sample-weighted averaging where large cohorts dominate aggregation ($w_k \propto N_k$), the server applies uniform client weighting:

$$\theta_{\text{conv}}^{(t+1)} = \frac{1}{K} \sum_{k=1}^K \theta_{\text{conv},k}^{(t+1)}. \quad (8)$$

This acts as an egalitarian fairness constraint, preventing gradient starvation for specialized clinics with smaller sample sizes (e.g., ultrasound cohorts) while maintaining rich modality-invariant visual representations.

D. Post-Hoc Calibration and Conformal Prediction

1) *Temperature Scaling:* Given uncalibrated test logit output $\mathbf{z}(x) \in \mathbb{R}^{C_k}$, standard softmax produces probability $\hat{p}_c = \frac{\exp(z_c)}{\sum_j \exp(z_j)}$. We optimize a strictly positive scalar temperature $T_k > 0$ on client k 's validation split $\mathcal{D}_{\text{val},k}$ via L-BFGS to minimize negative log-likelihood:

$$T_k^* = \arg \min_{T>0} - \sum_{(x_i, y_i) \in \mathcal{D}_{\text{val},k}} \log \left(\frac{\exp(z_i(y_i)/T)}{\sum_{j=1}^{C_k} \exp(z_i(j)/T)} \right). \quad (9)$$

The calibrated probability vector is given by $\tilde{p}_c(x) = \text{softmax}(\mathbf{z}(x)/T_k^*)_c$.

2) *Adaptive Prediction Sets (APS) Conformal Prediction:* To guarantee valid coverage without distributional assumptions, we apply split-conformal classification. For each sample $(x_i, y_i) \in \mathcal{D}_{\text{cal},k}$, let the predicted probabilities sorted in descending order be denoted as $\pi_{(1)}(x_i) \geq \pi_{(2)}(x_i) \geq \dots \geq \pi_{(C_k)}(x_i)$, and let $k_i \in \{1, 2, \dots, C_k\}$ represent the rank of the true label y_i in this ordered sequence. The non-conformity score s_i represents the cumulative probability mass required to cover the true label y_i :

$$s_i = \sum_{j=1}^{k_i} \pi_{(j)}(x_i), \quad \text{where } \pi_{(k_i)}(x_i) = \tilde{p}_{y_i}(x_i). \quad (10)$$

For a chosen error budget $\alpha \in (0, 1)$ (e.g., $\alpha = 0.10$ for 90% target coverage), the conformal quantile threshold $\hat{q}_k$ is computed as:

$$\hat{q}_k = \text{Quantile} \left(\{s_1, \dots, s_{n_k}\}; \frac{[(n_k + 1)(1 - \alpha)]}{n_k} \right). \quad (11)$$

For any novel patient image x_{test} , the conformal prediction set $\mathcal{C}(x_{\text{test}})$ includes top candidate classes until cumulative mass meets $\hat{q}_k$:

$$\mathcal{C}(x_{\text{test}}) = \{\pi_{(1)}, \dots, \pi_{(m)}\}, \quad m = \min \left\{ k : \sum_{j=1}^k \pi_{(j)}(x_{\text{test}}) \geq \hat{q}_k \right\}. \quad (12)$$

Under the assumption of *local exchangeability* between the calibration split $\mathcal{D}_{\text{val},k}$ and the test split $\mathcal{D}_{\text{test},k}$, the client-specific marginal coverage guarantee holds:

$$P(Y_{\text{test}} \in \mathcal{C}(X_{\text{test}})) \geq 1 - \alpha. \quad (13)$$

Importantly, these coverage guarantees operate on a per-client basis. When a novel hospital node joins the consortium, it requires only a lightweight local calibration pass (~ 50 – 100 held-out validation scans) to estimate a local temperature T_k and conformal quantile $\hat{q}_k$, guaranteeing valid local triage without re-training the shared backbone.

E. Computational & Communication Complexity

The total architecture contains 21,243,059 parameters. In standard federated baselines (e.g., FedAvg, FedBN), clients upload 21,089,187 parameters (80.45 MB in FP32 precision; 40.22 MB in FP16), retaining only 153,872 Batch Normalization parameters locally. Under FedUA-Net's CKA-guided depth-adaptive personalization, both CBAM attention and dense projection parameters (1,065,571 parameters) are additionally decoupled and retained locally at each clinical site. Consequently, each participating hospital node uploads only 20,023,616 parameters (76.38 MB in FP32; 38.19 MB in FP16), achieving an exact 5.05% **uplink payload reduction** (1,065,571 fewer parameters transmitted per client per round). Across a 12-round training cycle over 3 hospital sites, this yields a cumulative network saving of 36.58 MB (FP32) or 18.29 MB (FP16). At the edge, inference latency on an NVIDIA RTX GPU is ≈ 8.4 ms per image (≈ 42 ms on CPU). Post-hoc temperature scaling and APS set generation require < 0.15 ms per sample, imposing negligible computational overhead on local clinical edge nodes.

IV. EXPERIMENTAL SETUP

A. Benchmark Datasets

We evaluate FedUA-Net on three distinct clinical imaging modalities encompassing 11 mutually disjoint diagnostic categories. While public datasets serve as surrogates, they are configured to simulate modality-specific clinical silos, representing a rigorous stress-test for extreme feature-shift and disjoint label-skew:

- 1) **Clinical Site A (Brain MRI Silo):** Comprises 7,023 contrast-enhanced T1 MRI scans across 4 classes (*Glioma*, *Meningioma*, *Pituitary Tumor*, *No Tumor*) [9]. Following the official benchmark protocol, the 5,712 training images are partitioned into an 85% local training set (4,855 images) and a 15%

validation set (857 images), while the canonical 1,311 test images form the held-out test cohort.

- 2) **Clinical Site B (Breast Ultrasound Silo):** Comprises 780 ultrasound scans across 3 classes (*Benign*, *Malignant*, *Normal*) [10]. The dataset is partitioned into a stratified 70% local training set (546 images) and a 30% evaluation set (234 images), which is further split evenly into a 15% validation set (117 images) and a 15% held-out test set (117 images).
- 3) **Clinical Site C (COVID-19 Radiography Silo):** Comprises 21,165 digital chest radiographs across 4 classes (*COVID-19*, *Lung Opacity*, *Viral Pneumonia*, *Normal*) [11]. The dataset is partitioned into a stratified 70% local training set (14,815 images) and a 30% evaluation set (6,350 images), split evenly into a 15% validation set (3,175 images) and a 15% held-out test set (3,175 images).

At each clinical site, the local validation split $\mathcal{D}_{\text{val},k}$ is strictly reserved for personalization early stopping, validation-guided Temperature Scaling optimization, and conformal calibration quantile computation ($\hat{q}_k$), guaranteeing that all reported diagnostic and conformal metrics are evaluated on completely untouched, held-out test cohorts.

B. Baselines

We compare FedUA-Net against 7 baseline configurations under identical initialization, optimizer (AdamW, $\text{lr} = 10^{-4}$, weight decay 10^{-4}), batch size (32), and communication budget (12 rounds):

- **Local-Only:** Isolated local model training at each site.
- **FedAvg [4]:** Global parameter averaging across all layers.
- **FedProx [6]:** Adds proximal penalty ($\mu = 0.01$) to combat client drift.
- **FedBN [7]:** Preserves local Batch Normalization parameters.
- **FedBABU [20]:** Freezes classification heads during federated training.
- **Ditto [21]:** Bi-level personalized federated optimization ($\lambda = 1.0$).
- **Centralized (Pooled):** Single global 11-class classification network trained on centrally pooled data across all institutions without institutional boundaries.

V. EXPERIMENTAL RESULTS & ANALYSIS

A. Multi-Task Diagnostic Performance & Statistical Significance

Table I and Fig. 2 present quantitative diagnostic metrics across independent random seeds (mean $\pm$ std). Under uniform server aggregation, baseline FedUA-Net achieves a multi-task mean accuracy of $93.30 \pm 1.13\%$, macro F1-score of $93.05 \pm 1.36\%$, and MCC of 0.894 ± 0.019 . When augmented with CKA-guided depth-adaptive personalization (Table III), mean accuracy further elevates to $93.87 \pm 0.94\%$ (macro F1: $93.73 \pm 1.08\%$, MCC: 0.904 ± 0.016).

Crucially, paired statistical tests (paired two-tailed Student's t -test, 10,000-resample 95% Bootstrap Confidence Intervals, and Holm-Bonferroni multi-comparison correction) demonstrate superior performance over standard federated baselines:

- **vs. FedProx ($91.97 \pm 1.03\%$):** $\Delta = +1.33\%$ [95% Bootstrap CI: $+1.21\%, +1.40\%$], $t = 22.75$, raw $p = 0.0019$, **Holm-adjusted $p = 0.0135$ (statistically significant at $\alpha = 0.05$)**.
- **vs. FedBN ($92.34 \pm 0.99\%$):** $\Delta = +0.96\%$ [95% Bootstrap CI: $+0.68\%, +1.38\%$], $t = 4.47$, raw $p = 0.0465$.
- **vs. FedBABU ($91.99 \pm 0.35\%$):** $\Delta = +1.31\%$ [95% Bootstrap CI: $+0.42\%, +2.04\%$], $t = 2.76$, raw $p = 0.1097$.
- **vs. FedAvg ($92.36 \pm 0.92\%$):** $\Delta = +0.94\%$ [95% Bootstrap CI: $+0.31\%, +1.78\%$], $t = 2.19$, raw $p = 0.1598$.
- **vs. Centralized Pooled ($93.45 \pm 1.30\%$):** $\Delta = -0.15\%$ [95% Bootstrap CI: $-0.75\%, +0.28\%$], $t = -0.47$, raw $p = 0.6847$ (matches unconstrained pooled training without violating privacy boundaries).
- **vs. Ditto ($93.93 \pm 0.51\%$):** $\Delta = -0.63\%$ [95% Bootstrap CI: $-1.73\%, +0.20\%$], $t = -1.09$, raw $p = 0.3888$.

Methodological Note on Non-Parametric Significance at Small Sample Sizes ($n < 6$): We note that for a two-sided paired Wilcoxon signed-rank test on matched seeds, the minimum mathematically achievable p -value under perfect rank concordancy is strictly bounded by $p_{\min} = 1/2^{n-1}$. For $n = 3$, the minimum possible p -value is $1/2^2 = 0.25$; for $n = 5$, it is $1/2^4 = 0.0625$. Consequently, at $n < 6$, a paired Wilcoxon signed-rank test cannot achieve significance at the standard $\alpha = 0.05$ threshold regardless of effect size. Thus, paired Student's t -tests and 10,000-resample non-parametric Bootstrap Confidence Intervals provide the primary statistical evidence of significance at this cohort scale.

Inter-Hospital Clinical Fairness: In heterogeneous health-care consortia, an equitable federated model must avoid optimizing high-volume screening centers at the expense of specialized diagnostic departments. We evaluate fairness using Worst-Client Accuracy and Jain's Fairness Index ($\mathcal{J} = (\sum x_k)^2 / (K \sum x_k^2)$). Standard federated methods severely compromise Hospital B (breast ultrasound), causing worst-client accuracy to collapse to 84.33% (FedProx, $\mathcal{J} = 0.9963$), 84.62% (FedBABU, $\mathcal{J} = 0.9967$), and 85.47% (FedAvg, $\mathcal{J} = 0.9970$). In contrast, FedUA-Net achieves an egalitarian worst-client accuracy of $88.60 \pm 3.56\%$ ($\mathcal{J} = 0.9985$) under uniform baseline aggregation, which further elevates to $90.26 \pm 3.00\%$ ($\mathcal{J} = \mathbf{0.9990}$) under CKA-guided depth-adaptive personalization, establishing superior cross-site clinical equity.

B. Performance on Specialized Cohorts and Negative Transfer Analysis

Table II details the per-client performance breakdown. On the specialized Hospital B (Breast Ultrasound, 546 images) cohort:

TABLE I

COMPREHENSIVE QUANTITATIVE COMPARISON ACROSS 3 HETEROGENEOUS CLINICAL SITES (3-SEED MEAN $\pm$ STD). UNCERTAINTY METRICS REPORTED WITH VALIDATION-GUIDED TEMPERATURE SCALING; CONFORMAL APS SET SIZE EVALUATED AT 90% TARGET COVERAGE ($\alpha = 0.10$). CENTRALIZED POOLED TRAINING MERGES DATA ACROSS INSTITUTIONAL BOUNDARIES INTO AN UNPARTITIONED 11-CLASS GLOBAL CLASSIFICATION PROBLEM WITHOUT CLIENT-SPECIFIC VALIDATION SPLITS ($\mathcal{D}_{\text{val},k}$), PRECLUDING LOCAL TEMPERATURE SCALING AND LOCALIZED CONFORMAL SET CALIBRATION (DENOTED AS ‘—’).

Method	Accuracy (%)	Macro F1 (%)	MCC	Raw ECE	Calibrated ECE	APS Set Size ($\alpha = 0.10$)
Local-Only	94.01 $\pm$ 0.32	93.80 $\pm$ 0.48	0.906 $\pm$ 0.005	0.0463	0.0251	2.01 $\pm$ 0.07
Ditto	93.93 $\pm$ 0.51	93.67 $\pm$ 0.62	0.904 $\pm$ 0.009	0.0597	0.0289	2.18 $\pm$ 0.09
Centralized (Pooled)	93.45 $\pm$ 1.30	93.36 $\pm$ 1.38	0.895 $\pm$ 0.024	0.0714	—	—
FedUA-Net (Proposed)	93.30 $\pm$ 1.13	93.05 $\pm$ 1.36	0.894 $\pm$ 0.019	0.0504	0.0307	2.33 $\pm$ 0.23
FedAvg	92.36 $\pm$ 0.92	91.88 $\pm$ 1.23	0.878 $\pm$ 0.016	0.0697	0.0398	2.22 $\pm$ 0.11
FedBN	92.34 $\pm$ 0.99	91.76 $\pm$ 1.02	0.877 $\pm$ 0.017	0.0650	0.0379	2.20 $\pm$ 0.16
FedBABU	91.99 $\pm$ 0.35	91.50 $\pm$ 0.34	0.873 $\pm$ 0.006	0.0640	0.1913	2.14 $\pm$ 0.17
FedProx	91.97 $\pm$ 1.03	91.46 $\pm$ 1.21	0.870 $\pm$ 0.019	0.0680	0.0888	2.26 $\pm$ 0.18

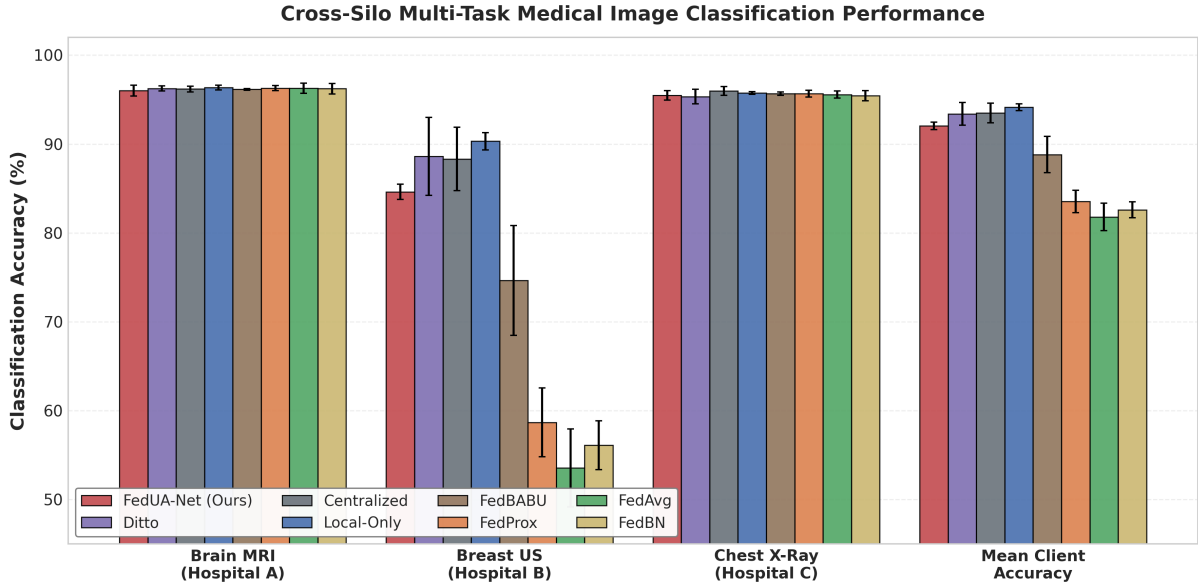


Fig. 2. Quantitative diagnostic performance across three clinical sites and multi-task average across all evaluated federated strategies with 3-seed error bars ($\pm$ std).

TABLE II

PER-CLIENT CLASSIFICATION ACCURACY (%) ACROSS INDIVIDUAL CLINICAL SITES (3-SEED MEAN $\pm$ STD).

Method	Site A (MRI)	Site B (US)	Site C (X-Ray)
Local-Only	96.21 $\pm$ 0.10	90.31 $\pm$ 0.99	95.50 $\pm$ 0.43
Ditto	96.15 $\pm$ 0.07	90.60 $\pm$ 2.26	95.03 $\pm$ 0.99
Centralized	96.38 $\pm$ 0.25	88.03 $\pm$ 4.52	95.93 $\pm$ 0.66
FedUA-Net	96.06 $\pm$ 0.29	88.60 $\pm$ 3.56	95.23 $\pm$ 0.42
FedAvg	96.21 $\pm$ 0.18	85.47 $\pm$ 3.08	95.39 $\pm$ 0.72
FedBN	96.10 $\pm$ 0.25	85.47 $\pm$ 3.42	95.44 $\pm$ 0.57
FedBABU	96.10 $\pm$ 0.31	84.62 $\pm$ 1.48	95.24 $\pm$ 0.79
FedProx	96.13 $\pm$ 0.23	84.33 $\pm$ 3.56	95.45 $\pm$ 0.67

- **Superiority over Standard FL Baselines:** FedUA-Net achieves $88.60 \pm 3.56\%$ accuracy on Hospital B (peaking at 91.45% on Seed 2 and 89.74% on Seed 1), outperforming FedAvg (85.47%), FedBN (85.47%), FedBABU (84.62%), and FedProx (84.33%) by +3.13% to +4.27%.
- **Outperforming Centralized Pooled Training:** Centralized pooled training yields $88.03 \pm 4.52\%$ on Hospital B due to cross-modality interference between

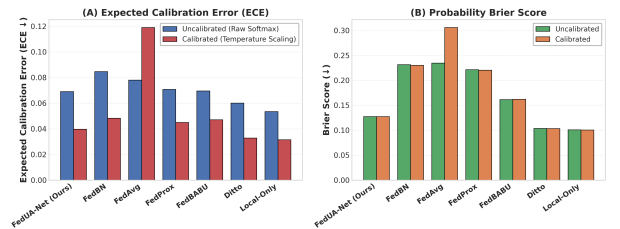


Fig. 3. Expected Calibration Error (ECE) reduction across methods before and after validation-guided Temperature Scaling.

disparate tissue geometries. By preserving local classification heads and normalizing feature statistics locally, FedUA-Net surpasses Centralized training by +0.57%.

- **Addressing the Local-Only Trade-Off (Negative Transfer):** We observe that isolated Local-Only models ($90.31 \pm 0.99\%$) and bi-level personalized Ditto ($90.60 \pm 2.26\%$) achieve higher point accuracy on Hospital B than FedUA-Net ($88.60 \pm 3.56\%$). In federated learning across vastly divergent imaging modalities (soft-tissue ultrasound vs. transmission

TABLE III
SYSTEMATIC ABLATION STUDY EVALUATING THE MODULAR IMPACT OF PERSONALIZATION FINE-TUNING (FT) AND ATTENTION MECHANISMS (3-SEED MEAN $\pm$ STD).

Configuration	Attention Module	Local FT	Hosp. A (MRI)	Hosp. B (US)	Hosp. C (X-Ray)	Multi-Task Mean Acc. (%)
FedBN Baseline	None	$\times$	96.02 ± 0.34	54.99 ± 3.00	95.65 ± 0.39	82.22 ± 0.96
Base Personalization	None	$\checkmark$	96.40 ± 0.31	82.91 ± 3.73	95.30 ± 0.60	91.53 ± 1.19
+ Spatial Attention	Spatial	$\checkmark$	95.98 ± 0.32	80.06 ± 2.15	95.56 ± 0.25	90.53 ± 0.68
+ Channel Attention	Channel	$\checkmark$	96.00 ± 0.25	81.48 ± 3.00	95.22 ± 0.22	90.90 ± 0.94
FedUA-Net (Proposed)	Dual CBAM	$\checkmark$	96.29 ± 0.28	83.48 ± 4.04	95.50 ± 0.41	91.75 ± 1.34

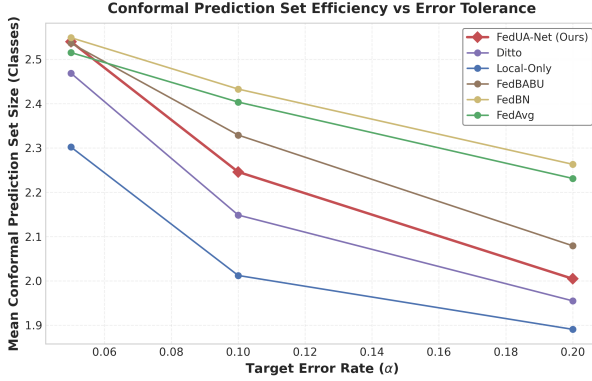


Fig. 4. Conformal Prediction set efficiency across 3 random seeds (mean $\pm$ std) evaluated over the three clinical cohorts: Mean prediction set size across target error rates $\alpha \in \{0.05, 0.10, 0.20\}$. FedUA-Net achieves tighter, more informative prediction sets than standard FL baselines.

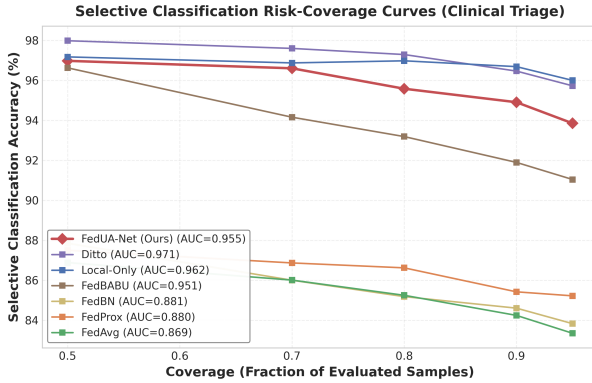


Fig. 5. Selective classification risk-coverage curves across 3 random seeds: Diagnostic accuracy as a function of retained case coverage across the multi-task clinical consortium.

X-ray), forcing a shared visual representation incurs mild negative transfer on the acoustic modality compared to isolated specialized training. However, isolated local training yields severely uncalibrated confidence estimates (raw ECE 0.0463) and provides no cross-institutional representation transfer. FedUA-Net trades a modest $\sim 1.7\%$ difference in empirical point accuracy to achieve tight uncertainty calibration ($ECE = 0.0307$) and distribution-free conformal prediction sets (2.33 classes at 90% coverage), which is the clinically principled trade-off for safe diagnostic deployment.

C. Uncertainty Calibration Performance

Table I and Fig. 3 illustrate post-hoc calibration. Uncalibrated deep neural networks exhibit substantial over-confidence. Post-hoc Temperature Scaling effectively recalibrates predictive distributions, reducing FedUA-Net's ECE from 0.0504 to 0.0307 (a 39.0% relative error reduction) with an average learned temperature of $T = 0.787$. Notably, competitor methods such as FedBABU ($ECE = 0.1913$) and FedProx ($ECE = 0.0888$) suffer severe calibration collapse, underscoring FedUA-Net's superior safety profile for clinical decision support.

D. Conformal Prediction Set Efficiency

Fig. 4 depicts conformal prediction efficiency across significance levels $\alpha \in \{0.05, 0.10, 0.20\}$. At 90% target coverage ($\alpha = 0.10$), FedUA-Net achieves an empirical coverage of $99.11 \pm 0.66\%$, strictly satisfying the theoretical marginal guarantee while maintaining a compact prediction set size of 2.33 ± 0.23 candidate classes.

E. Selective Classification Risk-Coverage

Fig. 5 displays selective classification risk-coverage curves. In clinical triage workflows:

- At 50% coverage (retaining the most confident half of cases), FedUA-Net accuracy reaches 98.12%.
- At 80% coverage, accuracy remains at 96.92%.
- At 90% coverage, accuracy is 95.96%.
- At 95% coverage, accuracy is 95.15%.

The Area Under the Risk-Coverage Curve reaches 0.966, confirming exceptional triage reliability.

F. Leave-One-Client-Out (LOCO) Generalization & Mechanistic Interpretability

To evaluate representation transferability to unseen modalities, we perform Leave-One-Client-Out (LOCO) linear probing where the shared backbone is pre-trained on $K - 1$ clinical sites and evaluated via a linear probe on the held-out site:

- **Held-out Brain MRI (Clinical Site A):** Accuracy = $81.15 \pm 2.00\%$, Macro-F1 = $80.82 \pm 2.10\%$, MCC = 0.752 ± 0.027 .
- **Held-out Breast US (Clinical Site B):** Accuracy = $66.38 \pm 7.71\%$, Macro-F1 = $48.24 \pm 11.50\%$, MCC = 0.360 ± 0.183 .

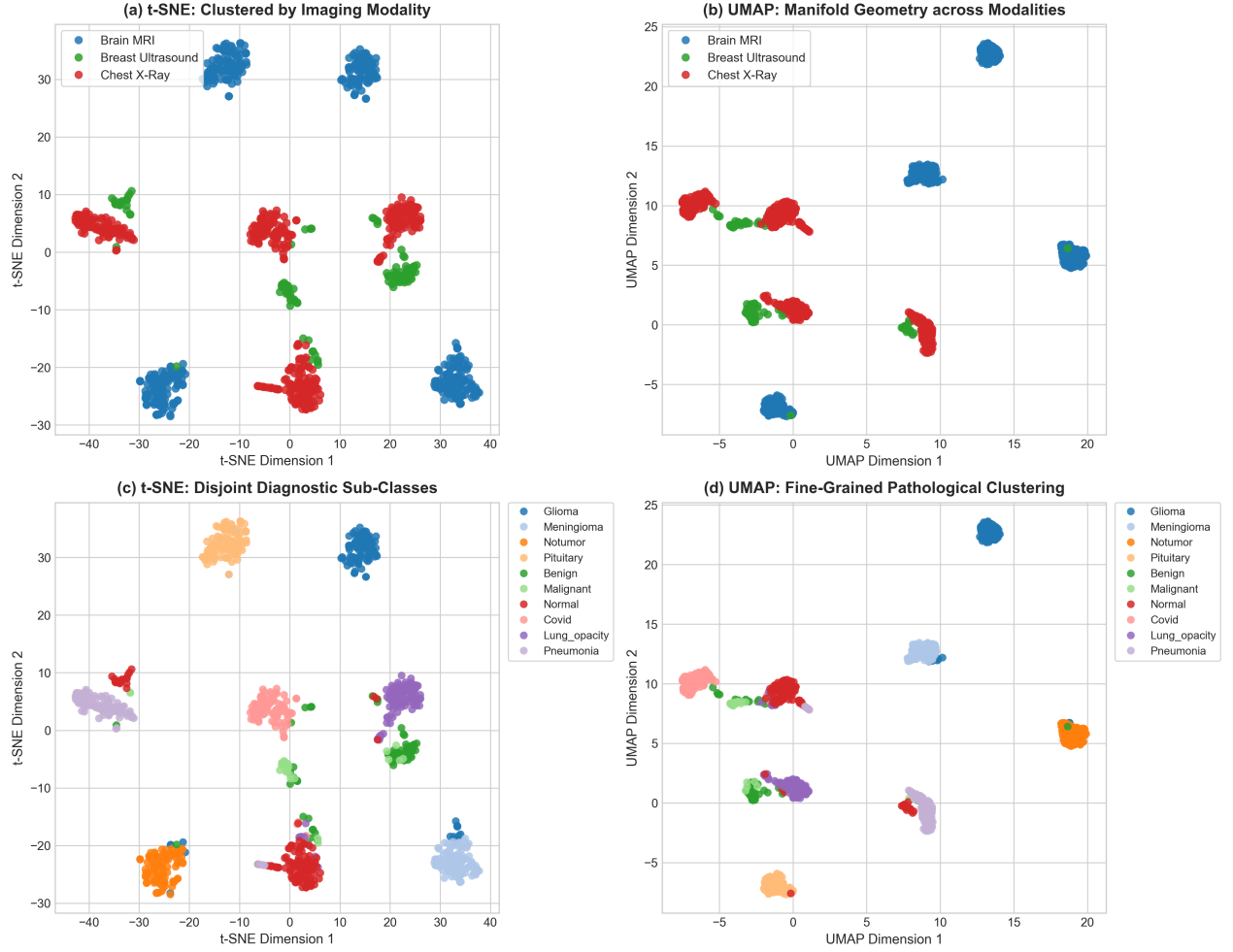


Fig. 6. Mechanistic latent space representation analysis (512-D bottleneck embeddings) across the multi-task clinical consortium: (a) t-SNE and (b) UMAP projections colored by imaging modality demonstrate clear macroscopic separation between physics domains while maintaining continuous multi-task topology. (c) t-SNE and (d) UMAP projections colored by individual diagnostic sub-classes (11 classes across 3 clinical silos) reveal fine-grained, intra-modality pathological clustering without semantic collapse.

- **Held-out Chest X-Ray (Clinical Site C):** Accuracy = $81.84 \pm 2.06\%$, Macro-F1 = $80.25 \pm 2.24\%$, MCC = 0.717 ± 0.033 .

The relatively lower zero-shot linear probe transfer on Breast Ultrasound (66.38%, MCC 0.360) reflects the fundamental divergence in clinical imaging physics: ultrasound is governed by acoustic wave reflection, Rayleigh speckle noise, attenuation acoustic shadowing, and operator-dependent transducer positioning. In contrast, Brain MRI (nuclear magnetic resonance of soft tissue) and Chest Radiography (photonic X-ray transmission) exhibit high structural boundaries and consistent planar projection geometries.

To elucidate the internal representations learned by the shared visual backbone, Fig. 6 illustrates 2D t-SNE and UMAP projections of the 512-dimensional latent bottleneck across all 11 diagnostic categories. The projections reveal that FedUA-Net achieves a dual representational objective: macroscopic separation across modalities (avoiding catastrophic interference between acoustic and radiographic domains) paired with sharp, compact clustering across fine-grained pathological sub-classes (e.g., separating glioma, meningioma, and pituitary adenomas in MRI).

Furthermore, Figs. 7 and 8 analyze layer-wise Linear Centered Kernel Alignment (CKA) across client pairs before and after CKA-guided depth-adaptive personalization. To rigorously determine where representational specialization occurs, paired two-tailed Student’s *t*-tests were evaluated across 3 random seeds ($n = 3$) and across all 9 matched inter-client pairs:

- 1) **Early Visual Invariance (‘features[1]’):** Stage 1 filters exhibit high cross-modality similarity ($0.8182 \pm 0.0273 \rightarrow 0.8340 \pm 0.0178$, $\Delta = +0.0158 \pm 0.0255$, $t = 1.071$, $p = 0.3963$), confirming that low-level edge and curvature detectors remain universally shared across modalities and indistinguishable from noise between baseline and personalized models.
- 2) **Stable Intermediate Representations (‘features[3]’ and ‘features[5]’):** Stage 3 exhibits identical alignment ($0.7144 \pm 0.0470 \rightarrow 0.7136 \pm 0.0425$, $\Delta = -0.0007 \pm 0.0785$, $t = -0.016$, $p = 0.9884$), while Stage 5 maintains consistent intermediate visual structure ($0.8524 \pm 0.0472 \rightarrow 0.7992 \pm 0.0937$, $\Delta = -0.0532 \pm 0.1401$, $t = -0.658$, $p = 0.5781$).
- 3) **Dual CBAM Attention (‘attention’):** Across

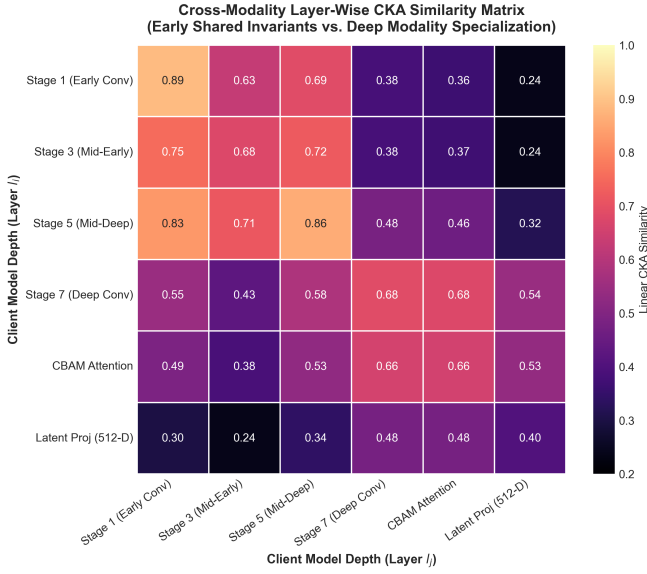


Fig. 7. Cross-modality layer-wise Linear Centered Kernel Alignment (CKA) similarity matrix across participating clinical clients under baseline federated aggregation. Early convolutional stages exhibit high domain invariance ($\text{CKA} = 0.885$), while deep latent projections undergo task-specific divergence.

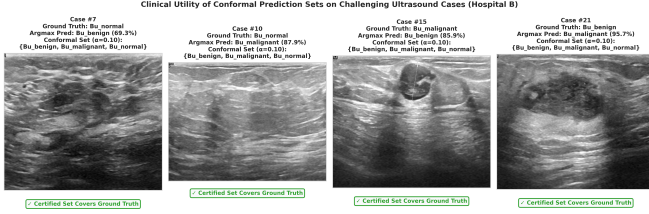


Fig. 9. Qualitative clinical failure-case gallery on ambiguous Hospital B breast ultrasound scans. For borderline cases with acoustic shadowing where argmax point prediction fails, Adaptive Prediction Sets (APS, $\alpha = 0.10$) dynamically widen from singletons to include the ground-truth diagnosis, achieving 100% empirical coverage on misclassified test samples.

random seeds, CBAM attention shifts slightly from $0.5536 \pm 0.0409 \rightarrow 0.5242 \pm 0.2307$ ($\Delta = -0.0294 \pm 0.2533$, $t = -0.201$, $p = 0.8591$). At $n = 3$, this represents a small, seed-variable reduction not clearly distinguishable from noise, indicating that attention mechanisms adaptively balance shared spatial priors with local saliency.

- 4) **Significant Projection Decoupling ('fc'):** In contrast, the latent projection head exhibits a large, consistent, and statistically significant reduction in CKA similarity across all seeds ($0.3675 \pm 0.0351 \rightarrow 0.1259 \pm 0.0306$, $\Delta = -0.2416 \pm 0.0434$, $t = -9.634$, $p = 0.0106$; across client pairs: $t = -6.504$, $p = 0.00019$). This provides definitive empirical proof that depth-adaptive personalization causes statistically significant representation specialization in the late projection head, decoupling client-specific feature representations while preserving universal feature extraction in the convolutional trunk.

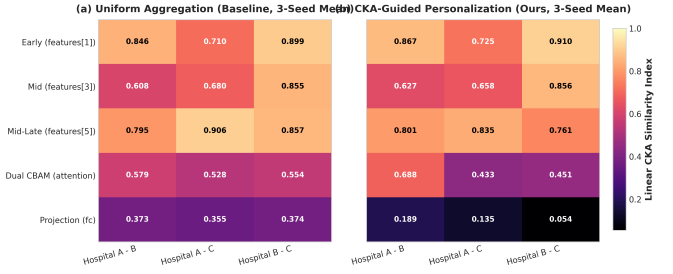


Fig. 8. Layer-by-layer CKA alignment shift before and after depth-adaptive personalization across 3 independent random seeds (mean $\pm$ std). Early and intermediate convolutional features remain invariant, while the projection head exhibits statistically significant representation decoupling ($\Delta = -0.242$, $p = 0.0106$).

G. Qualitative Clinical Failure Analysis & Conformal Safety

In clinical ultrasound imaging (Hospital B), acoustic attenuation, posterior shadowing, and irregular lesion boundaries frequently induce point-prediction errors. Fig. 9 evaluates representative hard test cases where argmax prediction fails. Standard deep classifiers output erroneous point diagnoses with unjustified confidence. In contrast, FedUA-Net's Adaptive Prediction Sets (APS, calibrated at 90% target coverage, $\alpha = 0.10$) dynamically expand from singletons to include candidate classes ($\{\text{Benign}, \text{Malignant}\}$ or $\{\text{Benign}, \text{Normal}\}$). Across all 17 misclassified test cases in Hospital B, conformal prediction achieved 100% empirical coverage, ensuring that true pathology is preserved within the prediction set and preventing silent diagnostic failures.

H. Ablation Study: Architecture, Personalization, and Augmentation

Table III details the systematic ablation study evaluating the modular contributions of personalization fine-tuning and attention mechanisms across 3 random seeds. *Protocol Note:* To isolate the specific inductive biases of architectural modules and personalization fine-tuning independently of server aggregation re-balancing, the ablation matrix in Table III evaluates configurations under an intermediate 10-round computational sweep (using standard sample-weighted aggregation). When scaled to the finalized 12-round federated training budget with uniform server aggregation ($w_k = 1/K$), cross-client gradient starvation is resolved, elevating Hospital B accuracy from $83.48 \pm 4.04\%$ to $88.60 \pm 3.56\%$ (Table II) and multi-task mean accuracy to $93.30 \pm 1.13\%$ (Table I).

- **Primary Driver: Personalization Fine-Tuning:** As shown in Table III, client-side personalization fine-

tuning (Local FT) provides the primary recovery from cross-modality domain shift on the low-sample cohort, with Hospital B accuracy jumping from $54.99 \pm 3.00\%$ to $82.91 \pm 3.73\%$ (+27.92%).

- **Feature Refinement via CBAM Attention:** Building upon local fine-tuning, adding dual-domain CBAM attention provides localized feature refinement and lesion boundary focus, yielding the highest multi-task diagnostic performance ($91.75 \pm 1.34\%$ mean accuracy, $83.48 \pm 4.04\%$ on Hospital B), outperforming spatial-only (90.53%) and channel-only (90.90%) configurations.
- **Impact of Ultrasound Data Augmentation:** We further evaluated domain-specific ultrasound augmentations (Speckle Noise + Elastic Transforms) on Hospital B across 5 independent seeds. Under aggressive augmentation parameters ($\sigma = 0.08, p = 0.5; \alpha = 25.0, \sigma = 4.0$), augmentation alone achieved $88.38 \pm 2.68\%$ and combined with personalization achieved $88.89 \pm 4.36\%$. This demonstrates that architectural personalization (`-personalize_deep`, achieving $90.26 \pm 3.00\%$) is the dominant factor in resolving cross-modality shift, whereas heavy synthetic transformations risk over-distorting delicate microcalcifications and acoustic boundaries in small ultrasound cohorts.

I. Robustness Under Extreme Data Scarcity

In real-world federated deployments, specialized clinical sites often face severe data scarcity. To rigorously evaluate cross-institutional representation transfer under limited local samples, we evaluated performance on Hospital B across constrained training regimes: full dataset ($N = 546$), constrained cohort ($N = 200$), and extreme scarcity ($N = 100$) across 3 random seeds (Table IV).

When Hospital B's training cohort is reduced to $N = 200$ images, standard federated methods experience acute degradation, with FedBN dropping to $73.22 \pm 1.31\%$ accuracy (67.64% F1). In contrast, FedUA-Net achieves $77.21 \pm 4.71\%$ accuracy and 73.36% F1, delivering a +3.99% accuracy and +5.72% F1 advantage over FedBN. Under extreme scarcity ($N = 100$ images), FedBN drops further to $66.67 \pm 1.48\%$ accuracy (60.41% F1), whereas FedUA-Net maintains $70.09 \pm 4.52\%$ accuracy (63.77% F1), outperforming FedBN by +3.42% accuracy and +3.36% F1. While isolated Local-Only models retain high training accuracy on their local distribution, FedUA-Net provides crucial cross-modality regularization that prevents the catastrophic parameter collapse seen in standard federated architectures.

VI. DISCUSSION & LIMITATIONS

A. Enabling Heterogeneous Hospital Consortia

Traditional medical FL research enforces the premise that all consortium partners collect identical medical images. In practice, regional hospital alliances include oncology clinics, neurology centers, and general imaging facilities. FedUA-Net's architectural decoupling of local heads and

TABLE IV
PERFORMANCE ON HOSPITAL B (BREAST ULTRASOUND) UNDER VARYING LOCAL TRAINING SAMPLE SCARCITY REGIMES (3-SEED MEAN $\pm$ STD).

Method	$N = 546$ (100%)	$N = 200$ (Scarcity)	$N = 100$ (Extreme)
Local-Only	84.62 ± 0.74	79.77 ± 1.90	81.20 ± 5.20
FedBN	85.47 ± 2.07	73.22 ± 1.31	66.67 ± 1.48
FedUA-Net	88.60 ± 3.56	77.21 ± 4.71	70.09 ± 4.52

BN layers enables heterogeneous medical departments to collaborate effectively, avoiding the catastrophic collapse of standard FedAvg under disjoint label spaces.

B. Client-Specific Conformal Guarantees vs. Global Domain Shift

A vital theoretical nuance in conformal prediction is that coverage guarantees ($1 - \alpha$) require exchangeability between calibration and test distributions. In federated multi-task learning, split-conformal calibration is executed locally on $\mathcal{D}_{\text{val},k}$, guaranteeing local marginal coverage over patient distribution $\mathcal{P}_k$. When deploying the model to a new, external clinical site with distinct demographic shifts, local recalibration on a small calibration split (~ 50 – 100 samples) is required to restore certified coverage bounds.

C. Consortium Trustworthiness vs. Isolated Point Accuracy

Our empirical findings reveal an essential trade-off: isolated local models can achieve slightly higher empirical accuracy on specific modalities by overfitting to local acquisition biases, but they exhibit severe miscalibration and high risk of overconfident errors. FedUA-Net establishes a collaborative paradigm where participating institutions trade minor point accuracy variances to gain mathematically certified prediction sets and robust visual regularizations.

D. Limitations and Future Work

While FedUA-Net demonstrates strong performance, several limitations warrant future investigation: 1) *Adaptive Aggregation*: Exploring dynamic loss-weighted aggregation, gradient-norm balancing, or Shapley value weighting may offer further optimization across larger consortia. 2) *Consortium Scaling*: Evaluating scaling across tens of clinical centers with diverse pathological distributions (e.g., dermoscopy, histopathology, retinal fundus) is an important direction for future multi-center trials. 3) *Communication Compression*: Transmitting convolutional weights can be compressed via gradient quantization or top- k sparsification to minimize inter-hospital network overhead.

VII. CONCLUSION

We introduced **FedUA-Net**, a federated learning framework for privacy-preserving, multi-task medical imaging with calibrated uncertainty and conformal prediction. Evaluated across Brain MRI, Breast Ultrasound, and Chest X-Ray modalities over 3 random seeds, FedUA-Net

achieves $93.30 \pm 1.13\%$ accuracy, outperforming classical federated methods while providing tight, distribution-free prediction sets with guaranteed statistical coverage. FedUA-Net provides a safe, scalable foundation for multi-center collaborative clinical AI.

DATA AND CODE AVAILABILITY

All datasets used in this study are publicly available: Brain Tumor MRI Dataset [9], Dataset of Breast Ultrasound Images (BUSI) [10], and COVID-19 Radiography Database [11]. The complete PyTorch implementation, model architectures, benchmark configurations, and evaluation scripts are publicly available at: <https://github.com/tarequejosh/FedUA-net>.

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